SERIAL COMMUNICATIONS

The 4-wire serial interface communicates in either the SPI or I^{2}C protocol. The interface affectively autodetects the format being used, requiring no configuration control to select the format.

The ADXL355 multifunction pins are referred to by a single function of the pin, for example, $\overline{CS}$, when only that function is relevant.

SPI PROTOCOL

Wire the ADXL355 for SPI communication as shown in the connection diagram in Figure 67. The SPI protocol timing is shown in Figure 69 to Figure 72. The timing scheme follows the clock polarity (CPOL) = 0 and clock phase (CPHA) = 0. The SPI clock speed ranges from 100 kHz to 10 MHz.

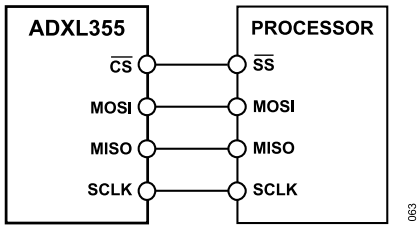

Figure 67. 4-Wire SPI Connection

SPI BUS SHARING

Use a gated buffer on the SCLK line for the ADXL355 device to achieve the ultralow noise performance and possibly offset shift when the ADXL355 must share a SPI bus with another subordinate device. This gated SCLK allows the clock signal through only when the chip select ($\overline{CS}$) line is low. See Figure 68 for the example circuit that provides this type of protection.

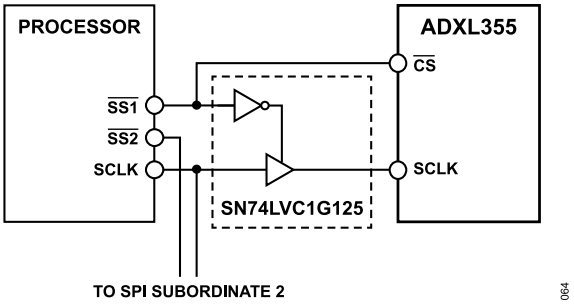

Figure 68. SCLK Protection Example

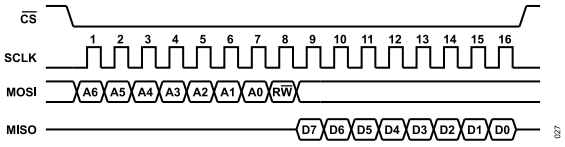

Figure 69. SPI Timing Diagram—Single-Byte Read

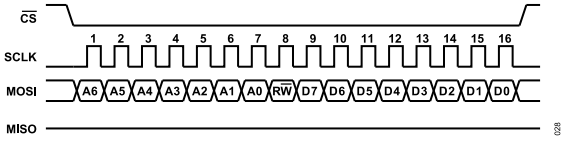

Figure 70. SPI Timing Diagram—Single-Byte Write

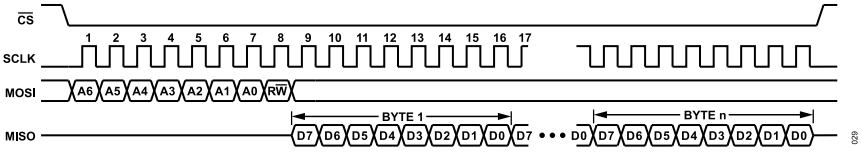

Figure 71. SPI Timing Diagram—Multibyte Read

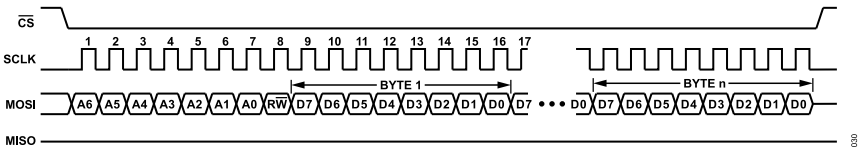

Figure 72. SPI Timing Diagram—Multibyte Write

SERIAL COMMUNICATIONS

I²C PROTOCOL

The ADXL355 supports point to point I²C communication. However, when sharing an SDA bus, the ADXL355 may prevent communication with other devices on that bus. If at any point, even when the ADXL355 is not being addressed, the 0x3A and 0x3B bytes (when the ADXL355 device address is set to 0x1D), or the 0xA6 and 0xA7 bytes (when the ADXL355 device address is set to 0x53) are transmitted on the SDA bus, the ADXL355 responds with an acknowledge bit and pulls the SDA line down. For example, this response can occur when reading or writing the data bytes (0x3A/0x3B or 0xA6/0xA7) to another sensor on the bus. When the ADXL355 pulls the SDA line down, communication with other devices on the bus may be interrupted. To resolve this interruption, the ADXL355 must be connected to a separate SDA bus, or the $\overline{CS}$/SCL pin must be switched high when communication with the ADXL355 is not desired (it is normally grounded).

The ADXL355 supports standard (100 kHz), fast (up to 1 MHz) and high speed (up to 3.4 MHz) data transfer modes when the bus parameters in Table 4 are met. There is no minimum SCL frequency, with the exception that, when reading data, the clock must be fast enough to read an entire sample set before new data overwrites it. Single-byte or multiple byte reads/writes are supported. With the MISO/ASEL pin low, the I²C address for the device is 0x1D and an alternate I²C address of 0x53 can be chosen by pulling the MISO/ASEL pin high.

There are no internal pull-up or pull-down resistors for any unused pins. Therefore, there is no known state or default state for the pins if left floating or unconnected. It is required that SCLK/V_{SSIO} be connected to ground when communicating to the ADXL355 using I²C.

Due to communication speed limitations, the maximum output data rate when using the 400 kHz I²C mode is 800 Hz, and it scales linearly with a change in the I²C communication speed. For example, using I²C at 100 kHz limits the maximum ODR to 200 Hz. Operation at an output data rate above the recommended maximum may result in an undesirable effect on the acceleration data, including missing samples or additional noise.

Figure 73 to Figure 75 detail the I²C protocol timing. The I²C interface can be used on most buses operating in I²C standard mode (100 kHz), fast mode (400 kHz), fast mode plus (1 MHz), and high speed mode (3.4 MHz). The ADXL355 I²C device ID is as follows:

- MISO/ASEL pin = 0, device address = 0x1D
- MISO/ASEL pin = 1, device address = 0x53

READING ACCELERATION OR TEMPERATURE DATA FROM THE INTERFACE

Acceleration data is left justified and has a register address order of most significant data to least significant data, which allows the user to use multibyte transfers and to take only as much data as required—8 bits, 16 bits, or 20 bits, plus the marker. Temperature data is 12 bits unsigned, right justified. The ADXL355 temperature value is split over two bytes, but is not double buffered, meaning the value can update between readings of the two registers. The data in XDATA, YDATA, and ZDATA is always the most recent available. It is not guaranteed that XDATA, YDATA, and ZDATA form a set corresponding to one sample point in time. The routine used to retrieve the data from the device controls this data set continuity. If data transfers are initiated when the DATA_RDY bit goes high and completes in a time approximately equal to 1/ODR, XDATA, YDATA, and ZDATA apply to the same data set.

For multibyte read or write transactions through either serial interface, the internal register address auto-increments. When the top of the register address range, 0x3FF, is reached, the auto-increment stops and does not wrap back to Address 0x00.

The address auto-increment function disables when the FIFO address is used, so that data can be read continuously from the FIFO as a multibyte transaction. In cases where the starting address of a multibyte transaction is less than the FIFO address, the address auto-increments until reaching the FIFO address, and then stops at the FIFO address.

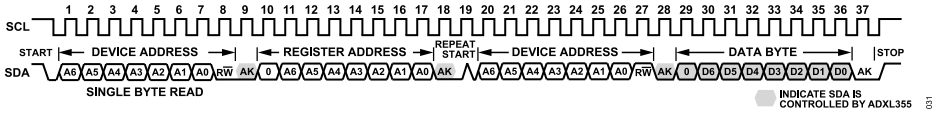

Figure 73. I²C Timing Diagram—Single-Byte Read

Figure 74. I²C Timing Diagram—Single-Byte Write

SERIAL COMMUNICATIONS

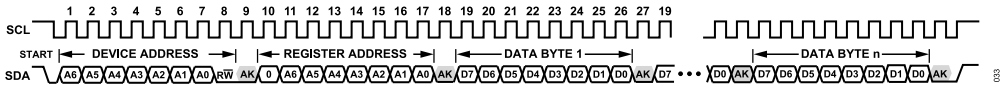

Figure 75. I²C Timing Diagram—Multibyte Write

FIFO

The FIFO operates in a stream mode. That is, when the FIFO overruns, new data overwrites the oldest data in the FIFO. A read from the FIFO address guarantees that the three bytes associated with the acceleration measurement on an axis all pertain to the same measurement. The FIFO never overflows, and the data is always taken out in sets (multiples of three data points).

There are 96 21-bit locations in the FIFO. Each location contains 20 bits of data and a marker bit for the x-axis data. A single-byte read from the FIFO address pops one location from the FIFO. A multibyte read to the FIFO location pops the FIFO on the read of the first byte and every third byte read thereafter.

Figure 76 shows the organization of the data in the FIFO. The acceleration data is twos complement, 20-bit data. The FIFO control logic inserts the two virtual bits (0b00) between the data bits and the empty indicator bit. Bit 1 indicates that an attempt was made to read an empty FIFO, and that the data is not valid acceleration data. Bit 0 is a marker bit to identify the x-axis, which allows a user to verify that the FIFO data was correctly read. An acceleration data point for a given axis occupies one FIFO location. The read pointer, RD_PTR, points to the oldest stored data that was not read already from the interface (see Figure 76). There are no physical x-acceleration, y-acceleration, or z-acceleration data registers. The data read from data registers (Register 0x08 to Register 0x10) also comes directly from the most recent data set in the FIFO, which is pointed to by the z pointer, Z_PTR, (see Figure 76).

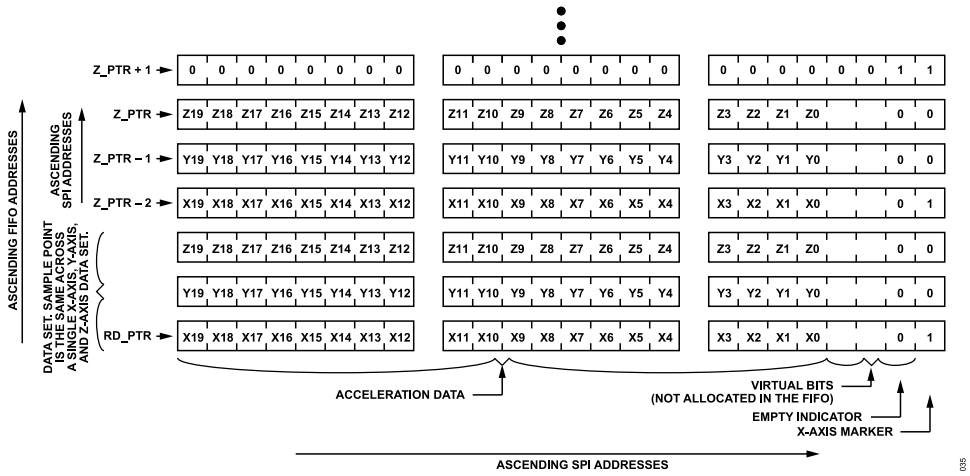

Figure 76. FIFO Data Organization

INTERRUPTS

The status register (Register 0x04) contains five individual bits, four of which can be mapped to the INT1 pin, the INT2 pin, or both. The polarity of the interrupt, active high or active low, is also selectable via the INT_POL bit in the range (Register 0x2C) register. In general, the status register clears when read, but this is not the case if the condition that caused the interrupt persists after the read of the register. The definition of persist varies slightly in each case, but it is described in the DATA_RDY, DRDY Pin, FIFO_FULL, FIFO_OVR, and Activity sections. The DRDY pin is similar to an interrupt pins (INTx) but clears differently. This case is also described.

DATA_RDY

The DATA_RDY bit is set when new acceleration data is available to the interface and clears on a read of the status register. This bit is not set again until acceleration data that is newer than the status register read is available.

Special logic on the clearing of the DATA_RDY bit covers the corner case where new data arrives during the read of the status register. In this case, the data ready condition may be missed completely. This logic results in a delay of the clearing of DATA_RDY of up to four 512 kHz cycles.

DRDY PIN

The DRDY pin is not a status register bit. DRDY instead behaves similar to an unmaskable interrupt. DRDY is set when new acceleration data is available to the interface. DRDY clears on a read of the FIFO, on a read of XDATA, YDATA, or ZDATA, or by an autoclear function that occurs approximately halfway between output acceleration data sets.

DRDY is always active high. The INT_POL bit does not affect DRDY. In external synchronization modes (EXT_SYNC = 01, EXT_SYNC = 10), the first few DRDY pulses after initial synchronization can be lost or corrupted. The length of this potential corruption is equal to or less than the group delay. Therefore, the samples within one group delay is lost or corrupted after the first synchronization signal. Depending on the decimation setting and interpolation setting (see Table 12), between one and three samples after the first synchronization pulse is lost, provided that all the restrictions set in the External Synchronization and Interpolation section is met.

FIFO_FULL

The FIFO_FULL bit is set when the entries in the FIFO are equal to the setting of the FIFO_SAMPLES bits. FIFO_FULL clears as follows:

- If the number of entries in the FIFO is less than the number of samples indicated by the FIFO_SAMPLES bits, which is only the case if sufficient data is read from the FIFO.
- On a read of the status register, but only when the entries in the FIFO are less than the FIFO_SAMPLES bits.

FIFO_OVR

The FIFO_OVR bit is set when the FIFO is so far overrange that data is lost. The specified size of the FIFO is 96 locations. The FIFO_OVR bit is set only when there is an attempt to write past this 96-location limit.

A read of the status register clears FIFO_OVR. FIFO_OVR is not set again until data is lost subsequent to this status register read.

ACTIVITY

The activity bit (Register 0x04, Bit 3) is set when the measured acceleration on any axis is above the value set in the ACT_ THRESH bits for ACT_COUNT consecutive measurements. An overthreshold condition can shift from one axis to another on successive measurements and is still counted toward the consecutive ACT_COUNT count.

A read of the status register clears the activity bit (Register 0x04, Bit 3), but the bit sets again at the end of the next measurement if the activity bit (Register 0x04, Bit 3) conditions are still satisfied.

NVM_BUSY

The NVM_BUSY bit indicates that the nonvolatile memory (NVM) controller is busy and, therefore, the NVM cannot be accessed to read or write. The interrupt functionality requires the NVM_BUSY bit to be cleared to function.

A status register read that occurs after the NVM controller is no longer busy clears NVM_BUSY.

EXTERNAL SYNCHRONIZATION AND INTERPOLATION

There are four possible synchronization options for the ADXL355, three of which are shown in Figure 77 to Figure 79. For clarity, the clock frequencies and delays are drawn to scale. The labels in Figure 77 to Figure 79 are defined as follows:

- Internal ODR is the alignment of the decimated output data based on the internal clock.
- ADC modulator clock shows the internal main clock rate.
- DRDY is an output indicator signaling a sample is ready.

The four possible synchronization options are as follows:

- No external synchronization (internal clocks used)
- Synchronization with an external synchronization signal and internal clock, interpolation filter enabled
- Synchronization with external synchronization and clock signals, no interpolation filter
- Synchronization with external synchronization and clock signals, interpolation filter enabled

INTERRUPTS

EXT_SYNC = 00, EXT_CLK = 0—No External Synchronization or Interpolation

This is the default mode of operation for the device. The sensor runs on an internal ODR and an internal clock that is generated by an internal oscillator. The internal ODR serves as the synchronization controller, which generates the data. Register 0x28 is used to program the ODR. No external signals are required, and this mode is used typically when the external processor retrieves data from the device asynchronously and absolute synchronization to an external source is not required.

The device outputs DRDY (active high) to signal that a new sample is available, and data is retrieved from the real-time registers or the FIFO. The group delay is based on the decimation setting, as shown in Table 10. This mode is shown in Figure 77.

EXT_SYNC = 10, EXT_CLK = 0—External Synchronization With Interpolation

Synchronization using interpolation filters and an external ODR clock is commonly used when the external processor can provide a synchronization signal that is asynchronous to the internal clock, SYNC, at the desired ODR. In this case, an interpolation filter provides additional time resolution of 64 times the programmed ODR (see Table 12). Synchronization with the interpolation filter enabled (EXT_SYNC = 10) allows the sensor to operate on an internal clock and output data most closely associated with the SYNC rising edge.

The advantage of this mode is that data is available at an arbitrary user defined SYNC sample rate and is asynchronous to the internal clock oscillator. The maximum sample rate cannot exceed 4000 SPS. The disadvantage of this mode is that the group delay is increased, with increased attenuation at the band edge. Additionally, because there is a limit to the time resolution, there is some distortion related to the mismatch of the external synchronization relative to the internal clock oscillator. This mismatch degrades spectral performance. The group delay is based on the decimation setting and interpolation setting (see Table 12). Figure 78 schematically shows the timings in this mode, and Table 13 shows the delay between the SYNC signal (input) and DRDY (output).

Table 13. EXT_SYNC = 10, DRDY Delay

ODR_LPF	SYNC to DRDY Delay (Oscillator Cycles)
0x0	8
0x1	10
0x2	14
0x3	22
0x4	38
0x5	70
0x6	134

Table 13. EXT_SYNC = 10, DRDY Delay (Continued)

ODR_LPF	SYNC to DRDY Delay (Oscillator Cycles)
0x7	262
0x8	1031
0x9	2054
0xA	4102

EXT_SYNC = 01, EXT_CLK = 1—External Synchronization and External Clock, No Interpolation Filter

When configured for EXT_SYNC = 01 and EXT_CLK = 1 (sync register, see Table 47), the user must supply an external clock (enabled via the EXT_CLK bit) at 1.024 MHz on the INT2 pin (Pin 13) and an external synchronization signal, SYNC, on the DRDY pin (Pin 14), as shown in Table 14. If configured in this mode and an external clock is not supplied, the device does not process any data and reading from the output results in null values. This mode is schematically shown in Figure 79.

Special restrictions when using this mode include the following:

- The external clock frequency on INT2 (Pin 13, see Table 14) must be 1.024 MHz.
- The pulse width of the SYNC signal must be at least 3.91 µs, which represents four cycles of the external clock (4 ÷ 1.024 MHz = ~3.91 µs).
- The phase of SYNC must meet an approximate 25 ns setup time to the external clock rising edge.

When using the EXT_SYNC mode and without providing the SYNC signal, the device runs on its own internal ODR. Similarly, after external synchronization, the device continues to run synchronized to the last SYNC pulse it received, which means that EXT_SYNC = 01 mode can be used with only a single synchronization pulse.

For more information about the lost sample in Figure 79, see the DRDY Pin section.

EXT_SYNC = 10, EXT_CLK = 1—External Synchronization and External Clock, With Interpolation Filter

This mode can be used to run the device on an external clock and synchronization with an arbitrary sample rate set by the SYNC signal rate. Conditions for external SYNC and external clock signals is the same as EXT_SYNC = 01, EXT_CLK = 1 mode. The interpolation filter provides a frequency resolution related to the ODR (see Table 12). In this case, the data provided corresponds to the external SYNC signal, which can be greater than the set ODR and less than 4000 SPS, but the output pass band remains the same it was prior to the interpolation filter.

INTERRUPTS

Table 14. Multiplexing of INT2 and DRDY

Register or Bit Fields			Pins		
EXT_CLK	EXT_SYNC, Bits[1:0]	INT_MAP, Bits[7:4]	INT2 (Pin 13)	DRDY (Pin 14)	Comments
0	00	0000	Low	DRDY	Synchronization is to the internal clocks, and there is no external clock synchronization.
0	00	Not 0000	INT2	DRDY	
1	00	0000	EXT_CLK	DRDY	
1	00	Not 0000[1]	EXT_CLK	DRDY	
0	01	0000	DRDY[2]	SYNC	These options reset the digital filters on every synchronization pulse and are not recommended.
0	01[3]	Not 0000	INT2	SYNC	
1	01[3]	0000	EXT_CLK	SYNC	External synchronization, no interpolation filter, and DRDY (active high) signals that data is ready. Data represents a sample point group delay earlier in time.
1	01[3]	Not 0000[1]	EXT_CLK	SYNC	
0	10	0000	DRDY[2]	SYNC	External synchronization, interpolation filter, and DRDY (active high) signals that data is ready. Data sample group delay earlier in time.
0	10[3]	Not 0000	INT2	SYNC	
1	10[3]	0000	EXT_CLK	SYNC	
1	10[3]	Not 0000	EXT_CLK	SYNC	

[1] No INT2, even though it is enabled.

[2] DRDY routing through the INT_MAP register takes precedence over the default, per Table 14.

[3] No DRDY.

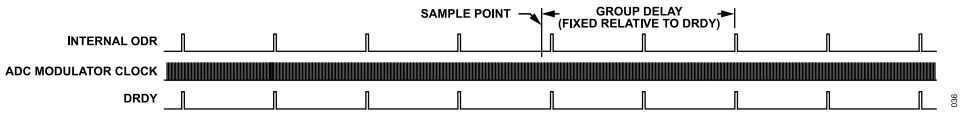

Figure 77. EXT_SYNC = 00, EXT_CLK = 0, Internal Synchronization, Internal Clock

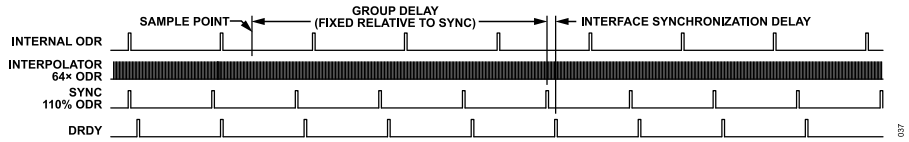

Figure 78. EXT_SYNC = 10, EXT_CLK = 0, External Synchronization, Internal Clock, Interpolation Filter

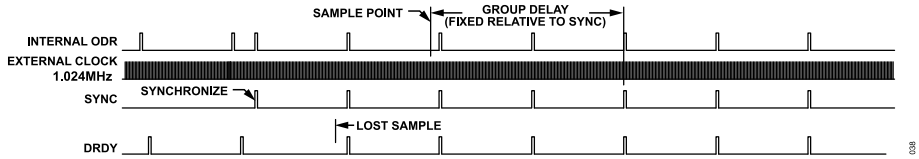

Figure 79. EXT_SYNC = 01, EXT_CLK = 1, External Synchronization, External Clock, No Interpolation Filter

ADXL355 REGISTER MAP

Note that while configuring the ADXL355 in an application, all configuration registers must be programmed before enabling measurement mode in the POWER_CTL register. When the ADXL355 is in measurement mode, only the following configurations can change: the HPF_CORNER bits in the filter register, the INT_MAP register, the ST1 and ST2 bits in the SELF_TEST register, and the reset register.

Table 15. ADXL355 Register Map

Hex. Addr.	Register Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Reset	R/W
0x00	DEVID_AD	DEVID_AD								0xAD	R
0x01	DEVID_MST	DEVID_MST								0x1D	R
0x02	PARTID	PARTID								0xED	R
0x03	REVID	REVID								0x01	R
0x04	Status	Reserved			NVM_BUSY	Activity	FIFO_OVR	FIFO_FULL	DATA_RDY	0x00	R
0x05	FIFO_ENTRIES	Reserved	FIFO_ENTRIES							0x00	R
0x06	TEMP2	Reserved				Temperature, Bits[11:8]				0x00	R
0x07	TEMP1	Temperature, Bits[7:0]								0x00	R
0x08	XDATA3	XDATA, Bits[19:12]								0x00	R
0x09	XDATA2	XDATA, Bits[11:4]								0x00	R
0x0A	XDATA1	XDATA, Bits[3:0]				Reserved				0x00	R
0x0B	YDATA3	YDATA, Bits[19:12]								0x00	R
0x0C	YDATA2	YDATA, Bits[11:4]								0x00	R
0x0D	YDATA1	YDATA, Bits[3:0]				Reserved				0x00	R
0x0E	ZDATA3	ZDATA, Bits[19:12]								0x00	R
0x0F	ZDATA2	ZDATA, Bits[11:4]								0x00	R
0x10	ZDATA1	ZDATA, Bits[3:0]				Reserved				0x00	R
0x11	FIFO_DATA	FIFO_DATA								0x00	R
0x1E	OFFSET_X_H	OFFSET_X, Bits[15:8]								0x00	R/W
0x1F	OFFSET_X_L	OFFSET_X, Bits[7:0]								0x00	R/W
0x20	OFFSET_Y_H	OFFSET_Y, Bits[15:8]								0x00	R/W
0x21	OFFSET_Y_L	OFFSET_Y, Bits[7:0]								0x00	R/W
0x22	OFFSET_Z_H	OFFSET_Z, Bits[15:8]								0x00	R/W
0x23	OFFSET_Z_L	OFFSET_Z, Bits[7:0]								0x00	R/W
0x24	ACT_EN	Reserved					ACT_Z	ACT_Y	ACT_X	0x00	R/W
0x25	ACT_THRESH_H	ACT_THRESH, Bits[15:8]								0x00	R/W
0x26	ACT_THRESH_L	ACT_THRESH, Bits[7:0]								0x00	R/W
0x27	ACT_COUNT	ACT_COUNT								0x01	R/W
0x28	Filter	Reserved		HPF_CORNER		ODR_LPF				0x00	R/W
0x29	FIFO_SAMPLES	Reserved	FIFO_SAMPLES							0x60	R/W
0x2A	INT_MAP	ACT_EN2	OVR_EN2	FULL_EN2	RDY_EN2	ACT_EN1	OVR_EN1	FULL_EN1	RDY_EN1	0x00	R/W
0x2B	Sync	Reserved					EXT_CLK	EXT_SYNC		0x00	R/W
0x2C	Range	I2C_HS	INT_POL	Reserved				Range		0x81	R/W
0x2D	POWER_CTL	Reserved					DRDY_OFF	TEMP_OFF	Standby	0x01	R/W
0x2E	SELF_TEST	Reserved						ST2	ST1	0x00	R/W
0x2F	Reset	Reset								0x00	W

REGISTER DEFINITIONS

This section describes the functions of the ADXL355 registers. The ADXL355 powers up with the default register values, as shown in the reset column of Table 15.

ANALOG DEVICES ID REGISTER

This register contains the Analog Devices ID, 0xAD.

Address: 0x00, Reset: 0xAD, Name: DEVID_AD

Table 16. Bit Descriptions for DEVID_AD

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	DEVID_AD		Analog Devices ID	0xAD	R

ANALOG DEVICES MEMS ID REGISTER

This register contains the Analog Devices MEMS ID, 0x1D.

Address: 0x01, Reset: 0x1D, Name: DEVID_MST

Table 17. Bit Descriptions for DEVID_MST

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	DEVID_MST		Analog Devices MEMS ID	0x1D	R

DEVICE ID REGISTER

This register contains the device ID, 0xED (355 octal).

Address: 0x02, Reset: 0xED, Name: PARTID

Table 18. Bit Descriptions for PARTID

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	PARTID		Device ID (355 octal)	0xED	R

PRODUCT REVISION ID REGISTER

This register contains the product revision ID, beginning with 0x00 and incrementing for each subsequent revision.

Address: 0x03, Reset: 0x01, Name: REVID

Table 19. Bit Descriptions for REVID

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	REVID		Mask revision	0x01	R

STATUS REGISTER

This register includes bits that describe the various conditions of the ADXL355.

Address: 0x04, Reset: 0x00, Name: Status

Table 20. Bit Descriptions for Status

Bits	Bit Name	Settings	Description	Reset	Access
[7:5]	Reserved		Reserved.	0x0	R
4	NVM_BUSY		NVM controller is busy with a refresh, programming, or a built in self test (BIST).	0x0	R

REGISTER DEFINITIONS

Table 20. Bit Descriptions for Status (Continued)

Bits	Bit Name	Settings	Description	Reset	Access
3	Activity		Activity, as defined in the ACT_THRESH_x and ACT_COUNT registers, is detected.	0x0	R
2	FIFO_OVR		FIFO has overrun, and the oldest data is lost.	0x0	R
1	FIFO_FULL		FIFO watermark is reached.	0x0	R
0	DATA_RDY		A complete x-axis, y-axis, and z-axis measurement was made and results can be read.	0x0	R

FIFO ENTRIES REGISTER

This register indicates the number of valid data samples present in the FIFO buffer. This number ranges from 0 to 96.

Address: 0x05, Reset: 0x00, Name: FIFO_ENTRIES

Table 21. Bit Descriptions for FIFO_ENTRIES

Bits	Bit Name	Settings	Description	Reset	Access
7	Reserved		Reserved	0x0	R
[6:0]	FIFO_ENTRIES		Number of data samples stored in the FIFO	0x0	R

TEMPERATURE DATA REGISTERS

These two registers contain the uncalibrated temperature data. The nominal intercept is 1885 LSB at 25°C and the nominal slope is −9.05 LSB/°C. TEMP2 contains the four most significant bits, and TEMP1 contains the eight least significant bits of the 12-bit value. The ADXL355 temperature value is not double buffered, meaning the value can update between reading of the two registers.

Address: 0x06, Reset: 0x00, Name: TEMP2

Table 22. Bit Descriptions for TEMP2

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	Reserved		Reserved.		
[3:0]	Temperature, Bits[11:8]		Uncalibrated temperature data	0x0	R

Address: 0x07, Reset: 0x00, Name: TEMP1

Table 23. Bit Descriptions for TEMP1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	Temperature, Bits[7:0]		Uncalibrated temperature data	0x00	R

X-AXIS DATA REGISTERS

These three registers contain the x-axis acceleration data. Data is left justified and formatted as twos complement.

Address: 0x08, Reset: 0x00, Name: XDATA3

Table 24. Bit Descriptions for XDATA3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	XDATA, Bits[19:12]		X-axis data	0x00	R

Address: 0x09, Reset: 0x00, Name: XDATA2

Table 25. Bit Descriptions for XDATA2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	XDATA, Bits[11:4]		X-axis data	0x00	R

REGISTER DEFINITIONS

Address: 0x0A, Reset: 0x00, Name: XDATA1

Table 26. Bit Descriptions for XDATA1

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	XDATA, Bits[3:0]		X-axis data	0x0	R
[3:0]	Reserved		Reserved	0x0	R

Y-AXIS DATA REGISTERS

These three registers contain the y-axis acceleration data. Data is left justified and formatted as twos complement.

Address: 0x0B, Reset: 0x00, Name: YDATA3

Table 27. Bit Descriptions for YDATA3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	YDATA, Bits[19:12]		Y-axis data	0x00	R

Address: 0x0C, Reset: 0x00, Name: YDATA2

Table 28. Bit Descriptions for YDATA2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	YDATA, Bits[11:4]		Y-axis data	0x00	R

Address: 0x0D, Reset: 0x00, Name: YDATA1

Table 29. Bit Descriptions for YDATA1

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	YDATA, Bits[3:0]		Y-axis data	0x0	R
[3:0]	Reserved		Reserved	0x0	R

Z-AXIS DATA REGISTERS

These three registers contain the z-axis acceleration data. Data is left justified and formatted as twos complement.

Address: 0x0E, Reset: 0x00, Name: ZDATA3

Table 30. Bit Descriptions for ZDATA3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	ZDATA, Bits[19:12]		Z-axis data	0x00	R

Address: 0x0F, Reset: 0x00, Name: ZDATA2

Table 31. Bit Descriptions for ZDATA2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	ZDATA, Bits[11:4]		Z-axis data	0x00	R

REGISTER DEFINITIONS

Address: 0x10, Reset: 0x00, Name: ZDATA1

Table 32. Bit Descriptions for ZDATA1

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	ZDATA, Bits[3:0]		Z-axis data	0x0	R
[3:0]	Reserved		Reserved	0x0	R

FIFO ACCESS REGISTER

Address: 0x11, Reset: 0x00, Name: FIFO_DATA

Read this register to access data stored in the FIFO.

Table 33. Bit Descriptions for FIFO_DATA

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FIFO_DATA		FIFO data is formatted to 24 bits, three bytes, most significant byte first. A read to this address pops an effective three equal byte words of axis data from the FIFO. Two subsequent reads or a multibyte read completes the transaction of this data onto the interface. Continued reading or a sustained multibyte read of this field continues to pop the FIFO every third byte. Multibyte reads to this address do not increment the address pointer. If this address is read due to an auto-increment from the previous address, it does not pop the FIFO. Instead, it returns zeros and increments on to the next address.	0x0	R

X-AXIS OFFSET TRIM REGISTERS

Address: 0x1E, Reset: 0x00, Name: OFFSET_X_H

Table 34. Bit Descriptions for OFFSET_X_H

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	OFFSET_X, Bits[15:8]		Offset added to x-axis data after all other signal processing. Data is in twos complement format. The significance of OFFSET_X, Bits[15:0] matches the significance of XDATA, Bits[19:4].	0x0	R/W

Address: 0x1F, Reset: 0x00, Name: OFFSET_X_L

Table 35. Bit Descriptions for OFFSET_X_L

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	OFFSET_X, Bits[7:0]		Offset added to x-axis data after all other signal processing. Data is in twos complement format. The significance of OFFSET_X, Bits[15:0] matches the significance of XDATA, Bits[19:4].	0x0	R/W

Y-AXIS OFFSET TRIM REGISTERS

Address: 0x20, Reset: 0x00, Name: OFFSET_Y_H

Table 36. Bit Descriptions for OFFSET_Y_H

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	OFFSET_Y, Bits[15:8]		Offset added to y-axis data after all other signal processing. Data is in twos complement format. The significance of OFFSET_Y, Bits[15:0] matches the significance of YDATA, Bits[19:4].	0x0	R/W

Address: 0x21, Reset: 0x00, Name: OFFSET_Y_L

Table 37. Bit Descriptions for OFFSET_Y_L

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	OFFSET_Y, Bits[7:0]		Offset added to y-axis data after all other signal processing. Data is in twos complement format. The significance of OFFSET_Y, Bits[15:0] matches the significance of YDATA, Bits[19:4].	0x0	R/W

REGISTER DEFINITIONS

Z-AXIS OFFSET TRIM REGISTERS

Address: 0x22, Reset: 0x00, Name: OFFSET_Z_H

Table 38. Bit Descriptions for OFFSET_Z_H

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	OFFSET_Z, Bits[15:8]		Offset added to z-axis data after all other signal processing. Data is in twos complement format. The significance of OFFSET_Z, Bits[15:0] matches the significance of ZDATA, Bits[19:4].	0x0	R/W

Address: 0x23, Reset: 0x00, Name: OFFSET_Z_L

Table 39. Bit Descriptions for OFFSET_Z_L

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	OFFSET_Z, Bits[7:0]		Offset added to z-axis data after all other signal processing. Data is in twos complement format. The significance of OFFSET_Z, Bits[15:0] matches the significance of ZDATA, Bits[19:4].	0x0	R/W

ACTIVITY ENABLE REGISTER

Address: 0x24, Reset: 0x00, Name: ACT_EN

Table 40. Bit Descriptions for ACT_EN

Bits	Bit Name	Settings	Description	Reset	Access
[7:3]	Reserved		Reserved.	0x0	R
2	ACT_Z		Z-axis data is a component of the activity detection algorithm.	0x0	R/W
1	ACT_Y		Y-axis data is a component of the activity detection algorithm.	0x0	R/W
0	ACT_X		X-axis data is a component of the activity detection algorithm.	0x0	R/W

ACTIVITY THRESHOLD REGISTERS

Address: 0x25, Reset: 0x00, Name: ACT_THRESH_H

Table 41. Bit Descriptions for ACT_THRESH_H

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	ACT_THRESH, Bits[15:8]		Threshold for activity detection. Acceleration magnitude must be above ACT_THRESH to trigger the activity counter. ACT_THRESH is an unsigned magnitude. The significance of ACT_THRESH, Bits[15:0] matches the significance of Bits[18:3] of XDATA, YDATA, and ZDATA.	0x0	R/W

Address: 0x26, Reset: 0x00, Name: ACT_THRESH_L

Table 42. Bit Descriptions for ACT_THRESH_L

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	ACT_THRESH, Bits[7:0]		Threshold for activity detection. The acceleration magnitude must be greater than the value in ACT_THRESH to trigger the activity counter. ACT_THRESH is an unsigned magnitude. The significance of ACT_THRESH, Bits[15:0] matches the significance of Bits[18:3] of XDATA, YDATA, and ZDATA.	0x0	R/W

REGISTER DEFINITIONS

ACTIVITY COUNT REGISTER

Address: 0x27, Reset: 0x01, Name: ACT_COUNT

Table 43. Bit Descriptions for ACT_COUNT

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	ACT_COUNT		Number of consecutive events above threshold (from ACT_THRESH) required to detect activity	0x1	R/W

FILTER SETTINGS REGISTER

Address: 0x28, Reset: 0x00, Name: Filter

Use this register to specify parameters for the internal high-pass and low-pass filters.

Table 44. Bit Descriptions for Filter

Bits	Bit Name	Settings	Description	Reset	Access
7	Reserved		Reserved	0x0	R
[6:4]	HPF_CORNER		−3 dB filter corner for the first-order, high-pass filter relative to the ODR	0x0	R/W
		000	Not applicable, no high-pass filter enabled		
		001	$24.7 \times 10^{-4} \times$ ODR		
		010	$6.2084 \times 10^{-4} \times$ ODR		
		011	$1.5545 \times 10^{-4} \times$ ODR		
		100	$0.3862 \times 10^{-4} \times$ ODR		
		101	$0.0954 \times 10^{-4} \times$ ODR		
		110	$0.0238 \times 10^{-4} \times$ ODR		
[3:0]	ODR_LPF		ODR and low-pass filter corner	0x0	R/W
		0000	4000 Hz and 1000 Hz		
		0001	2000 Hz and 500 Hz		
		0010	1000 Hz and 250 Hz		
		0011	500 Hz and 125 Hz		
		0100	250 Hz and 62.5 Hz		
		0101	125 Hz and 31.25 Hz		
		0110	62.5 Hz and 15.625 Hz		
		0111	31.25 Hz and 7.813 Hz		
		1000	15.625 Hz and 3.906 Hz		
		1001	7.813 Hz and 1.953 Hz		
		1010	3.906 Hz and 0.977 Hz		

FIFO SAMPLES REGISTER

Address: 0x29, Reset: 0x60, Name: FIFO_SAMPLES

Use the FIFO_SAMPLES value to specify the number of samples to store in the FIFO. The default value of this register is 0x60 to avoid triggering the FIFO watermark interrupt.

Table 45. Bit Descriptions for FIFO_SAMPLES

Bits	Bit Name	Settings	Description	Reset	Access
7	Reserved		Reserved.	0x0	R
[6:0]	FIFO_SAMPLES		Watermark number of samples stored in the FIFO that triggers a FIFO_FULL condition. Values range from 1 to 96.	0x60	R/W

REGISTER DEFINITIONS

INTERRUPT PIN (INTX) FUNCTION MAP REGISTER

Address: 0x2A, Reset: 0x00, Name: INT_MAP

The INT_MAP register configures the interrupt pins. Bits[7:0] select which functions generate an interrupt on the INT1 and INT2 pins. Multiple events can be configured. If the corresponding bit is set to 1, the function generates an interrupt on the interrupt pins.

Table 46. Bit Descriptions for INT_MAP

Bits	Bit Name	Settings	Description	Reset	Access
7	ACT_EN2		Activity interrupt enable on INT2	0x0	R/W
6	OVR_EN2		FIFO_OVR interrupt enable on INT2	0x0	R/W
5	FULL_EN2		FIFO_FULL interrupt enable on INT2	0x0	R/W
4	RDY_EN2		DATA_RDY interrupt enable on INT2	0x0	R/W
3	ACT_EN1		Activity interrupt enable on INT1	0x0	R/W
2	OVR_EN1		FIFO_OVR interrupt enable on INT1	0x0	R/W
1	FULL_EN1		FIFO_FULL interrupt enable on INT1	0x0	R/W
0	RDY_EN1		DATA_RDY interrupt enable on INT1	0x0	R/W

DATA SYNCHRONIZATION

Address: 0x2B, Reset: 0x00, Name: Sync

Use this register to control the external timing triggers.

Table 47. Bit Descriptions for Sync

Bits	Bit Name	Settings	Description	Reset	Access
[7:3]	Reserved		Reserved.	0x0	R
2	EXT_CLK		Enable external clock. See Table 14 for configuration details.	0x0	R/W
[1:0]	EXT_SYNC		Enable external synchronization control.	0x0	R/W
		00	Internal synchronization.		
		01	External synchronization, no interpolation filter. After synchronization, and for EXT_SYNC within specification, DATA_RDY occurs on EXT_SYNC.		
		10	External synchronization, interpolation filter, next available data indicated by DATA_RDY 14 to 8204 oscillator cycles later (longer delay for higher ODR_LPF setting), data represents a sample point group delay earlier in time.		
		11	Reserved.		

I²C SPEED, INTERRUPT POLARITY, AND RANGE REGISTER

Address: 0x2C, Reset: 0x81, Name: Range

Table 48. Bit Descriptions for Range

Bits	Bit Name	Settings	Description	Reset	Access
7	I2C_HS		I²C speed.	0x1	R/W
		1	High speed mode.		
		0	Fast mode.		
6	INT_POL		Interrupt polarity.	0x0	R/W
		0	INT1 and INT2 are active low.		
		1	INT1 and INT2 are active high.		
[5:2]	Reserved		Reserved.	0x0	R
[1:0]	Range		Range.	0x1	R/W
		01	±2 *g*.		

REGISTER DEFINITIONS

Table 48. Bit Descriptions for Range (Continued)

Bits	Bit Name	Settings	Description	Reset	Access
		10	±4 *g*.		
		11	±8 *g*.		

POWER CONTROL REGISTER

Address: 0x2D, Reset: 0x01, Name: POWER_CTL

Table 49. Bit Descriptions for POWER_CTL

Bits	Bit Name	Settings	Description	Reset	Access
[7:3]	Reserved		Reserved.	0x0	R
2	DRDY_OFF		Set to 1 to force the DRDY output to 0 in modes where it is normally signal data ready.	0x0	R/W
1	TEMP_OFF		Set to 1 to disable temperature processing. Temperature processing is also disabled when standby = 1.	0x0	R/W
0	Standby		Standby or measurement mode.	0x1	R/W
		1	Standby mode. In standby mode, the device is in a low power state, and the temperature and acceleration datapaths are not operating. In addition, digital functions, including FIFO pointers, reset. Changes to the configuration setting of the device must be made when standby = 1. An exception is a high-pass filter that can be changed when the device is operating.		
		0	Measurement mode.		

SELF TEST REGISTER

Address: 0x2E, Reset: 0x00, Name: SELF_TEST

Refer to the Self Test section for more information on the operation of the self test feature.

Table 50. Bit Descriptions for SELF_TEST

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	Reserved		Reserved.	0x0	R
1	ST2		Set to 1 to enable self test force	0x0	R/W
0	ST1		Set to 1 to enable self test mode	0x0	R/W

RESET REGISTER

Address: 0x2F, Reset: 0x00, Name: Reset

Table 51. Bit Descriptions for Reset

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	Reset		Write Code 0x52 to reset the device, similar to a power-on reset (POR)	0x0	W

In case of a software reset, an unlikely race condition may occur in products with REVID = 0x01 or earlier. If the race condition occurs, some factory settings in the NVM load incorrectly to shadow registers (the registers from which the internal logic configures the sensor and calculates the output after a power-on or a software reset). The incorrect loading of the NVM affects overall performance of the sensor, such as an incorrect *0 g* bias and other performance issues. The incorrect loading of NVM does not occur from a power-on or after a power cycle. To guarantee reliable operation of the sensor after a software reset, the user can access the shadow registers after a power-on, read and store the values on the host microprocessor, and compare the values read from the same shadow registers after a software reset. This method guarantees proper operation in all devices and under all conditions. The recommended steps are as follows:

1. Read the shadow registers, Register 0x50 to Register 0x54 (five 8-bit registers) after power-up, but before any software reset.
2. Store these values in a host device (for example, a host microprocessor).
3. After each software reset, read the same five registers. If the values differ, perform a software reset again until they match.